

Hudson Schmidt

Park City, UT | San Luis Obispo, CA

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Education

California Polytechnic State University, San Luis Obispo

Bachelor of Science in Computer Science, Emphasis in Cybersecurity

Expected June 2027

Relevant Coursework: Wireless Security; Database Systems; Computer Security; Privacy: Policy and Technology; Distributed Computing

Planned Coursework: Network Security; Networks; System Security; Operating Systems

Research Interests

Cybersecurity, wireless and network security, privacy, and dependable software systems.

Research Experience

Student Research Assistant, FuTURES³ Lab, University of Utah

Salt Lake City, UT | June 2025 - September 2025

- Parallelized OGHarn's Python harness generator across CPU cores, reducing runtime by approximately 67% while achieving roughly 2x higher code coverage on target C libraries.
- Designed experiments and benchmarks to evaluate coverage and runtime trade-offs and guide multiprocessing parameters and defaults.
- Collaborated with a Ph.D. mentor to document methodology and integrate improvements into the research codebase.

Technical Projects

Homebound | Personal Project

September 2025 - Present

- Developing a SwiftUI iOS application with a Python FastAPI backend that enables users to create safety plans with automated check-ins and emergency contact notifications for overdue arrivals.
- Building end-to-end authentication flows, push notification systems, and real-time status tracking to support user safety and trip accountability with a focus on privacy.
- Publicly released the application on the iOS App Store in 35 countries.
- Technologies: Python, Swift, SQL, HTML/CSS

App Store: www.apple.co/4qNNCn9

Project Website: www.homeboundapp.com

Haptic Feedback System for Visually Impaired Marching Band | Empower Club at Cal Poly

December 2024 – Present

- Prototyping an ESP32 and ESP-NOW wireless system that provides real-time haptic guidance for 26 musicians.
- Overseeing code development, hardware prototyping, and iterative testing to improve system reliability and user experience.
- Technologies: ESP32, ESP-NOW, Python, C++

GitHub: <https://github.com/hudsonschmidt/HaptiBand>

Recent Presentation: <https://tinyurl.com/4wvrp6vd>

Stein Eriksen Collection Website Redesign | PCCAPS Project

January 2023 – May 2023

- Delivered wireframes and a clickable prototype for a redesigned website for a premier Deer Valley resort group.
- Contributed to a modernized web presence that informed the final site design.

Technical Skills

Programming Languages: *Proficient:* Python, SQL. *Exposure to:* C, Swift, Java, JavaScript, HTML/CSS.

Tools and Frameworks: *Proficient:* Git, Linux CLI, VS Code, Supabase/Postgres, Wireshark.

Exposure to: Docker, React, Node.

Professional Experience

Orientation Leader | Cal Poly New Student & Transition Programs | San Luis Obispo, CA

April 2024 – September 2024

- Led onboarding for 24 first-year students through full-day orientation programming and resource coordination.
- Facilitated debriefs and Q&A sessions to help students navigate academic expectations, campus resources, and university life.

Lead Host & Head Runner | Blind Dog Restaurant | Park City, UT

January 2023 – Present (Seasonal)

- Support front-of-house coordination during peak service by managing hosting, food running, garnishing, and delivery workflows.
- Assist with training new team members across multiple operational roles, contributing to efficient service and strong guest experiences.

Leadership and Service

Volunteer Diver and Underwater Surveyor | Padma Organization and Conservation Diver

- Supported biodiversity surveys and reef protection initiatives in the Azores (2022) and Bali/Komodo (2023).

Lion's Heart Teen Volunteers & Leaders | 2016 – 2023

- Participated in sustained community service involving food security, elder care, animal rescue, and environmental cleanups.